

Interview Guide and Talking Points

Sector Reversal Rotation - explained from first principles

The 30-second explanation

I built a rules-based strategy that compares the recent performance of 11 stock-market sectors. Every 20 trading days, it buys the three sectors that performed worst over the prior 20 trading days. The idea is short-term mean reversion: sectors can become temporarily oversold and then bounce back. I included a one-day execution lag, transaction costs, portfolio weights, and standard risk metrics so the backtest is reproducible.

Explain it like I am new to finance

- A sector ETF is a basket of companies from one part of the economy, such as technology or energy.
- Return means percentage price change. A stock moving from \$100 to \$105 has a 5% return.
- Ranking means sorting sectors from best to worst.
- Mean reversion means unusually weak performance may partially reverse toward normal.
- Equal weighting means splitting the money evenly among the selected sectors.
- A backtest applies the rule to old data as though the decisions were made at that time.

Why the one-day lag matters

Without a lag, code can accidentally use a closing price to create a position that earns that same day's return. That is impossible in real trading. I calculate the signal at day t and apply the weight starting on day $t+1$.

Key mathematics in plain English

- 20-day return = current adjusted price divided by the price 20 trading days ago, minus one.
- Portfolio return = weighted average of the selected sector returns.
- Turnover = how much the portfolio weights change when rebalancing.
- Transaction cost = turnover multiplied by 0.10%.
- Sharpe ratio = return per unit of volatility; higher is better, but it can be inflated by a short or overfit sample.

- Maximum drawdown = the worst percentage fall from a previous portfolio peak.

Results you can state carefully

Metric	Strategy	SPY
Total Return	39.6%	31.1%
CAGR	39.6%	31.1%
Annualized Volatility	10.5%	12.5%
Sharpe (rf=0)	3.22	2.23
Max Drawdown	-3.6%	-6.6%
Positive Days	53.2%	59.1%

Use this wording: “On the included 2019 in-sample dataset, the selected specification returned 39.6% versus 31.1% for SPY and had a maximum drawdown of 3.6% versus 6.6%. Because I selected parameters after examining the sample, I treat this as hypothesis generation, not proof of an investable edge.”

Likely interview questions

Why mean reversion instead of momentum?

I tested relative sector behavior and found that short-horizon laggards reversed in this sample. Economically, broad sectors can overshoot after flows or news, but I would test both reversal and momentum out of sample.

Is this look-ahead biased?

The trade implementation is not same-bar look-ahead biased because weights are shifted one day. However, the project deliberately contains parameter-selection bias because the final lookback and portfolio size were selected in sample.

Why use adjusted prices?

Adjusted prices account for distributions and splits, making historical returns more comparable.

Why 20 days?

It approximates one trading month. It was also selected from a parameter search, which is why longer out-of-sample testing is essential.

What would you do next?

Use walk-forward validation over 2005-present data, stress transaction costs, test nearby parameters, compare alternative benchmarks, and report confidence intervals.

Would you trade this with real money?

Not based on this result alone. I would require out-of-sample stability, paper trading, execution analysis, and explicit risk limits.

What not to claim

- Do not say the model guarantees profits.
- Do not call the 2019 result out-of-sample.
- Do not hide that parameters were selected after testing alternatives.
- Do not imply the high Sharpe ratio is stable across decades.

Strong resume bullets

- Developed an event-driven Python backtester for an 11-sector ETF rotation strategy with lagged signals, transaction costs, allocation records, and automated performance reports.
- Implemented cross-sectional mean-reversion ranking and evaluated CAGR, volatility, Sharpe ratio, turnover, and maximum drawdown against SPY.
- Documented model risk, multiple-testing bias, and a walk-forward validation roadmap to distinguish research performance from deployable alpha.

Code walkthrough

- `load_prices()`: reads the included CSV or downloads Yahoo Finance data.
- `backtest()`: calculates trailing returns, ranks sectors, builds weights, shifts execution, and subtracts costs.
- `stats()`: calculates return and risk metrics.
- `save_outputs()`: exports CSVs and charts for auditability.